
Analytical System for Agriculture Region Grouping

Problem Id: 04

Student Name : ISHITA BIJALWAN

University Roll number : 2028434

Section: U

Student Name : SUHANI RAWAT

University Roll number : 2028504

Section: U

Student Name : HRITAMBRA

University Roll number : 2028430

Section: U

1.Problem Description:

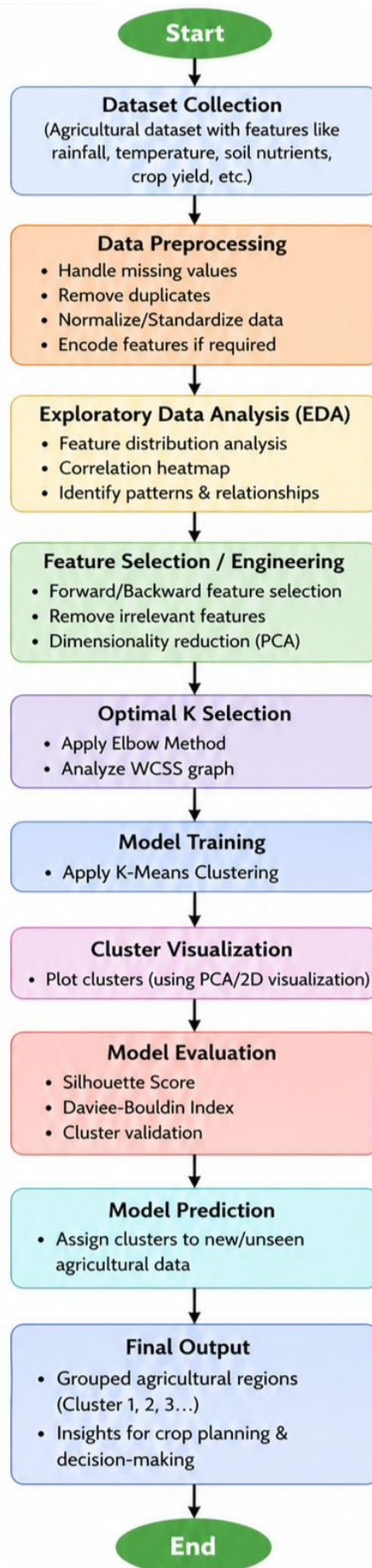
Agriculture varies widely across regions due to differences in environmental conditions and resource availability. Factors like rainfall, temperature, soil quality, and farming practices directly impact crop productivity. However, analyzing these multiple factors together to identify meaningful patterns is difficult using traditional methods.

This project aims to solve that problem by developing a data-driven system using K-Means clustering. The goal is to group agricultural regions based on similarities in crop yield and environmental characteristics. By doing this, regions with similar conditions and productivity patterns can be identified.

The system will first analyze the dataset to understand distributions and relationships between features. Then, it will clean and normalize the data to ensure fair comparison. After selecting the most relevant features, clustering will be applied to divide regions into meaningful groups.

Finally, the results will help in better understanding agricultural patterns, improving decision-making, and predicting cluster categories for new regions. This makes the system useful for planning, resource allocation, and improving crop yield strategies.

2. Architecture



3. Models Used(Applied/ Planned) :

- ***K-Means Clustering***

This is the main model used in the project. It groups agricultural regions into clusters based on similarity in features like rainfall, soil nutrients, temperature, etc. The algorithm works by dividing data into K groups where each group has similar characteristics. It helps in identifying patterns between regions with similar crop yield conditions.

- ***Elbow Method***

This method is used to find the optimal number of clusters (K). It checks how the clustering performance changes as K increases and helps choose a value where improvement slows down. Basically, it prevents choosing too many or too few clusters.

- ***Silhouette Score***

This is used to evaluate how good the clustering is. It measures how similar a data point is to its own cluster compared to other clusters. A higher score means better clustering quality.

- ***Principal Component Analysis (PCA) (Planned)***

PCA is used for dimensionality reduction. Since the dataset has many features, PCA helps reduce them into fewer dimensions while keeping important information. It also helps in visualization of clusters in 2D or 3D space.

- ***Feature Selection Techniques (Forward/Backward Elimination) (Planned)***

These methods help in selecting only the most important features that contribute to clustering. This improves model performance and reduces unnecessary complexity.

- ***Data Preprocessing (Normalization / Standardization)***

Before applying clustering, features are scaled using techniques like normalization or standardization. This ensures all features (like rainfall, temperature, etc.) are on the same scale so no single feature dominates the clustering.

4.Evaluation metrics used:

- ***Silhouette Score***

This is the main metric used to evaluate clustering quality. It measures how well each data point fits within its cluster compared to other clusters.

Value ranges from -1 to 1
Closer to 1 = good clustering
Around 0 = overlapping clusters
Negative = wrong clustering

- ***Within-Cluster Sum of Squares (WCSS)***

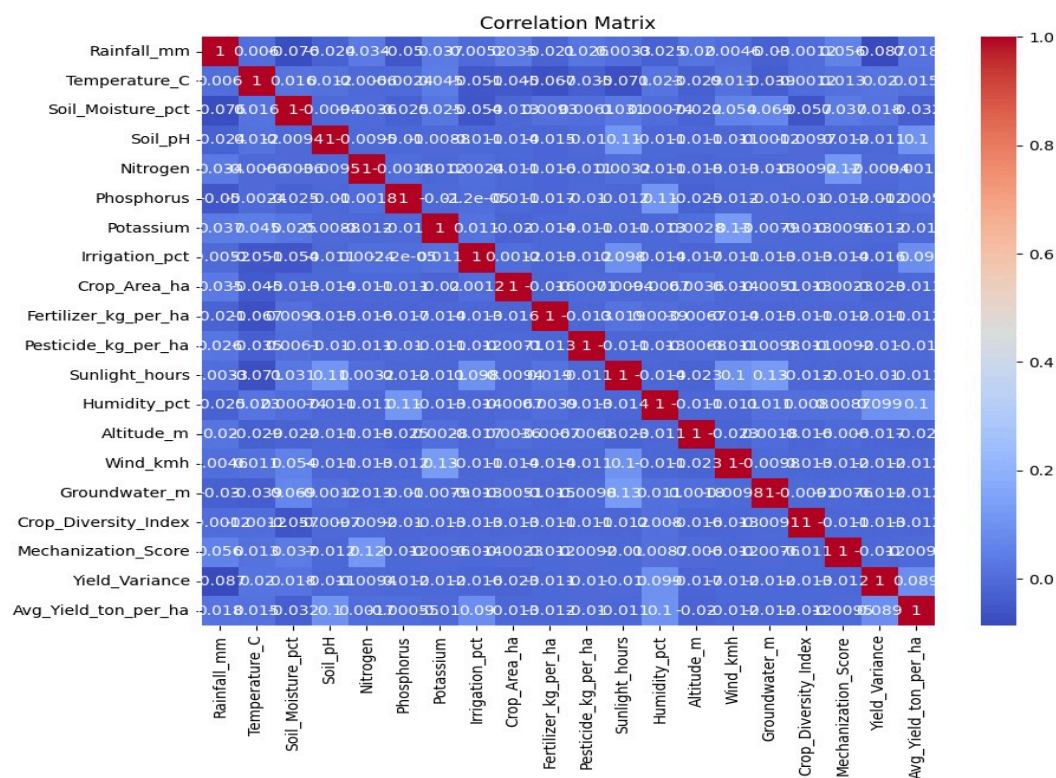
This measures how compact the clusters are. It calculates the distance between data points and their cluster centroid.

Lower WCSS = tighter and better clusters

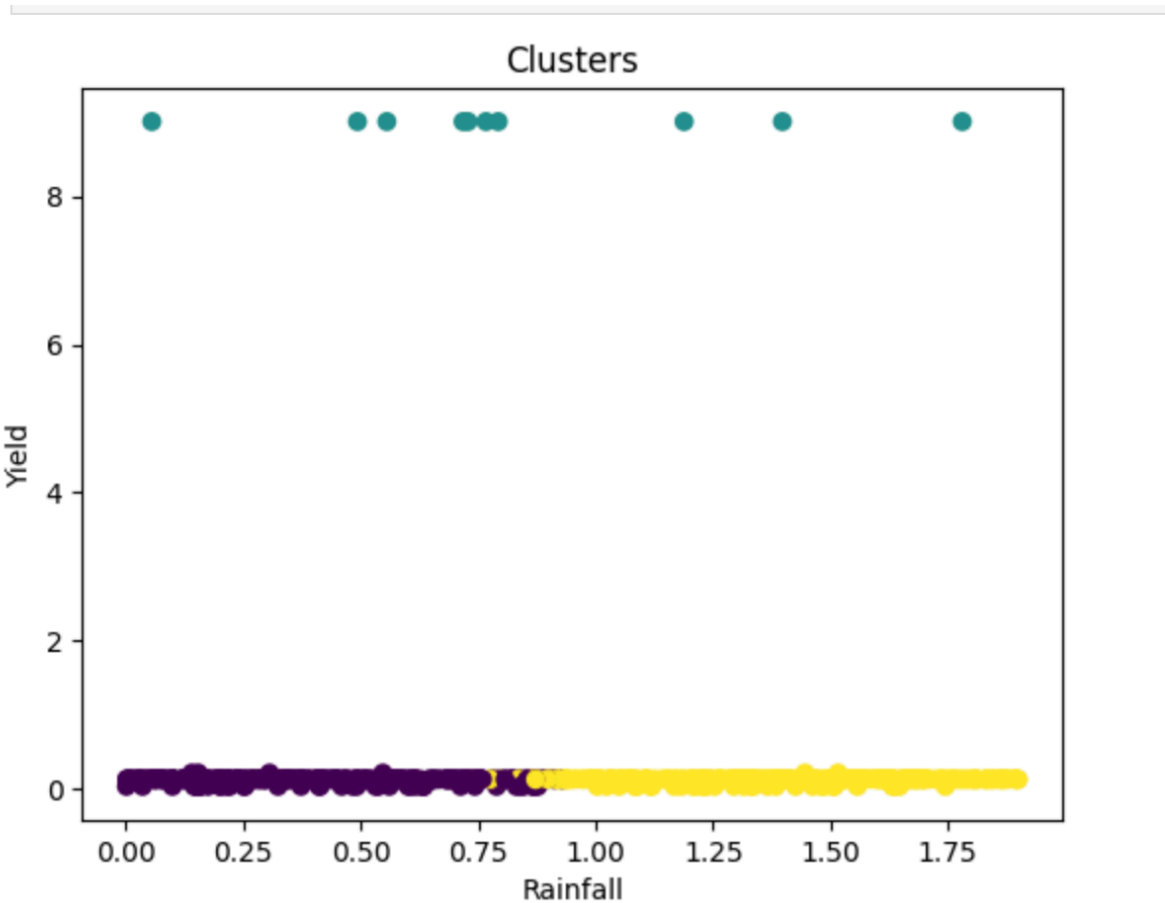
- ***Cluster Visualization***

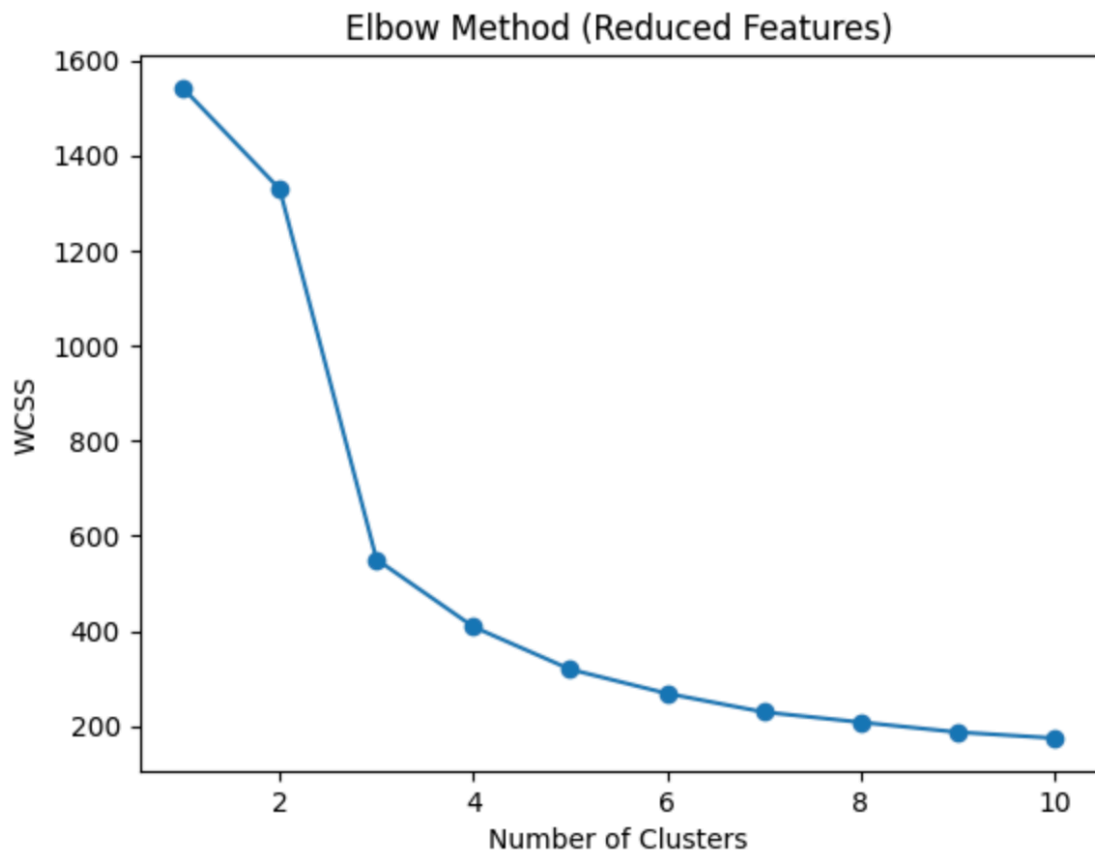
Graphs (like PCA plots) are used to visually check if clusters are clearly separated.

5. Preliminary Results and Visualizations



Heatmap showing correlation matrix





6. References:

Books:

- Pattern Recognition and Machine Learning – Christopher M. Bishop
- Introduction to Machine Learning with Python – Andreas C. Müller

Online Resources:

- Scikit-learn – <https://scikit-learn.org>
- GeeksforGeeks – <https://www.geeksforgeeks.org>
- TutorialsPoint – <https://www.tutorialspoint.com>
- Stack Overflow – <https://stackoverflow.com>